Ductal carcinoma in situ of the breast: finding the balance between overtreatment and undertreatment

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Ductal carcinoma in situ (DCIS) accounts for 15–25% of all breast cancer diagnoses. Its prognosis is excellent overall, the main risk being the occurrence of local breast events, as most cases of DCIS do not progress to invasive cancer. Systematic screening has greatly increased the incidence of this non-obligate precursor of invasion, lending urgency to the need to identify DCIS that is prone to invasive progression and distinguish it from non-invasion-prone DCIS, as the latter can be overdiagnosed and therefore overtreated. Treatment strategies, including surgery, radiotherapy, and optional endocrine therapy, decrease the risk of local events, but have no effect on survival outcomes. Active surveillance is being evaluated as a possible new option for low-risk DCIS. Considerable efforts to decipher the biology of DCIS have led to a better understanding of the factors that determine its variable natural history. Given this variability, shared decision making regarding optimal, personalised treatment strategies is the most appropriate course of action. Well designed, risk-based de-escalation studies remain a major need in this field.

Introduction

Ductal carcinoma in situ (DCIS) is regarded as "a neoplastic proliferation of cells within the ductal-lobular structures of the breast that has not penetrated the myoepithelial basement membrane interface."[1] It can progress to invasive breast cancer, but in many cases, it never does.[2] As such, DCIS is a non-obligate invasive precursor to breast cancer. Based on autopsy findings, its prevalence could be over 20% among women older than 40 years.[3] Pure DCIS is very rare among male individuals.[4] This Review therefore focuses on data for female individuals.

Before the implementation of mammographic breast cancer screening, DCIS was a rare finding, but since then, its incidence has increased manifold.[5] Approximately 25% of all screen-detected breast cancers are DCIS.[6] Even among populations where screening is well established, a large reservoir of cases of undiscovered DCIS remains.[3] Yet, no concurrent decline in the incidence of more advanced stages of invasive breast cancer has been observed, suggesting that a diagnosis of DCIS might represent an overdiagnosis—a situation in which a neoplastic pathology is diagnosed, which would have remained unnoticed in the absence of screening.[7] In a large retrospective US-based study published in 2019,[8] only 10–15% of women who did not receive any treatment after the diagnosis of a DCIS developed invasive cancer at 5·5 years median follow-up. Modelling studies estimate that up to 65% of DCIS might be overdiagnosed depending on whether spontaneous regression is allowed.[2] This uncertain natural history, in which atypical lesions and DCIS appear as non-obligate precursors of invasive cancer (ie, invasive breast cancer can occur either from a DCIS lesion or directly), unfortunately limits the positive effect of current breast cancer screening, unlike in screening for cervical and colorectal cancer.

For many years, DCIS has been treated aggressively with surgery, radiotherapy, and endocrine therapy, yet the long-term risk of dying of breast cancer in women diagnosed with DCIS remains very low at approximately 3%.[9,10] Additionally, an analysis of data from the Surveillance, Epidemiology, and End Results (SEER) Program for 1286 people with treated DCIS[8] showed a 10-year risk of ipsilateral invasive breast cancer of 12·2% for grade 1 or 2 DCIS and a risk of 17·6% for grade 3 DCIS. Active surveillance is an increasingly common approach for many conditions with similar behaviour, including some atypical lesions of the breast,[11] Barret's oesophagus,[12] some renal cell cancers,[13] and low-risk prostate and thyroid tumours.[14] Active surveillance is therefore an important potential management option for DCIS but requires reliable and reproducible identification of low-risk lesions, together with prospective demonstration of the safety of

Search strategy and selection criteria

Studies were identified through searches of MEDLINE, Current Contents, PubMed, and references from relevant articles with the search terms "ductal carcinoma in situ", "breast", "DCIS" and "intraductal carcinoma", from 1990 to Dec 5, 2023. We considered peer-reviewed articles published in English or for which an English abstract was available. We selected all original publications of high quality, including all randomised trials, cohort studies, case control studies, well conducted reviews and meta-analyses, good quality epidemiological and real-world evidence studies, high-level preclinical studies published in reference journals, and other original or more exploratory publications. Management guidelines were identified through a search of MEDLINE and the websites of major relevant international cancer societies (the American Society of Clinical Oncology, European Society of Clinical Oncology, American Association for Cancer Research, European Society for Radiotherapy and Oncology, American Society for Radiation Oncology, and Society of Surgical Oncology). We identified past and ongoing clinical trials through searches of ClinicalTrials.gov and MEDLINE with the keywords "DCIS" and "breast" and "clinical trial". All authors had access to the full text of all selected publications. The relevant selected references were shared and discussed between all four authors over several rounds until a final consensus was reached. Given the large number of eligible publications, the final selection of references for this Review was based on the most recent and relevant publications with the highest combined level of evidence. The final content of this Review was obtained through repeated consensual discussion between all authors to limit bias. This Review might still be subject to bias, and this bias mainly reflects the authors' interpretation of current knowledge in this field.

this strategy. Pathology, biology, and genomic investigations have taken a great leap forward in the past 5 years to inform our understanding of the oncogenic continuum between breast precursor lesions (including DCIS) and invasive cancer.[15–22] This Review summarises the key knowledge available regarding the natural history of DCIS, prognostic markers, and treatment results to inform balanced management strategies that physicians should discuss with individuals diagnosed with this condition.

Understanding the natural history of DCIS

Risk factors predicting the development of DCIS do not differ from those of invasive breast cancer.[23] Germline genetic conditions that predispose to breast cancer, including *BRCA1* and *BRCA2*, also lead to an increased risk of DCIS,[24] whereas *TP53* alterations are associated with a very high risk of breast cancer of which 50% of cases are DCIS.[25] Men with DCIS and women with early onset DCIS (up to age 50 years)[26] or a family history should, therefore, be referred for genetic counselling.[25,26]

In the past 10 years, data have increased our knowledge on how and how often DCIS progresses to invasive cancer. The genetic concordance between DCIS and matched invasive breast cancer is high.[27] Four complementary models have been proposed.[20] The independent lineage model proposes that in situ and invasive cell populations arise from different cell lineages.[28] The convergent phenotype model proposes that different DCIS genotypes can ultimately progress to invasive breast cancer of the same phenotype, suggesting that progression to invasive breast cancer is driven by specific clones leading to the same phenotype.[29] The evolutionary bottleneck model assumes that individual ductal cells accumulate heterogeneous genetic aberrations, but only a cell subpopulation with a specific genetic profile is able to drive invasion.[15,29] In the multiclonal invasion model, multiple clones can escape from the ducts and comigrate into adjacent tissues to establish invasive carcinomas.[15,16] Data published in the past 3 years seem to corroborate the coexistence of the different models, although the independent lineage model appears to be rare. In a study of 129 primary DCIS matched to their respective local recurrences, of which 95 were invasive, a pangenomic analysis revealed that 75% (71 of 95) of invasive tumours were clonally related to the primary DCIS. 18% (17 of 95) were unrelated and 7% (7 of 95) were ambiguous.[18] Clonal relatedness was unrelated to time to recurrence, grade, or subtype. Genomic evolution occurred at very early stages of progression in ducts. These data suggest that DCIS might be either a precursor or a risk lesion for invasive breast cancer.[18] Another recent study that used Arc-well, a high-throughput single-cell DNA-sequencing method on archival formalin-fixed paraffin-embedded samples, showed the coexistence of different models, with a predominance of the evolutionary bottleneck model.[30] Finally, a phylogenetic analysis[31] has recently demonstrated the unique evolutionary histories of breast tissues, with driver mutations occurring in breast cells from healthy epithelia very early in life, and multiple independent DCIS or invasive breast cancer founders commonly arising from non-cancerous but clonally mutated ancestor cells.[31] How and why do some of these indolent lesions become invasive? In the past 5 years, data have highlighted the role of both cell-autonomous and non-autonomous events, with host-related factors and the microenvironment contributing to cancer promotion through the oncogenic switch from breast precursor lesions to invasion.[20,22,31] The transition to invasive breast cancer could be associated with progressive changes in the stromal microenvironment, as identified with a spatial imaging atlas of patient-matched DCIS and invasive breast cancers.[19] For breast epithelial cells, oncogenic driver mutations in cells from healthy breast epithelia have been shown to occur and might show clonal expansion in the ductal structures. Yet, to what extent these genomic and non-genomic events might drive DCIS initiation or potential progression to invasive breast cancer remains unknown.

Current challenges in the detection and diagnosis of DCIS: too much or too little?

Screening mammography is the driver of DCIS detection, where it typically presents as calcifications. Image-guided, 7–12-gauge biopsies provide accurate evaluation and, together with imaging-based lesion delineation, inform therapeutic decision making.[32] Mammography has evolved over the past 25 years, from film-screen studies to digital, and then tomosynthesis imaging.[33] Each step led to increased sensitivity, particularly for premenopausal women, but also to a potential increase in overdiagnoses.[34]

MRI is increasingly used for screening and surveillance among women at high risk of breast cancer, and individuals with mammographically dense breasts, as per recent international guidelines.[35,36] Consequently, the number of DCIS lesions being diagnosed during MRI screening examinations is increasing.[35,37] However, most of the information about MRI performance in DCIS comes from studies of post-diagnostic MRI, for which the goal is assessment of disease extent and there is an expected decrease in reoperation rates. In the 2021 multicentre, single-arm trial ECOG-ACRIN 4112, designed to evaluate preoperative MRI in 368 women with DCIS that was diagnosed on mammography and was eligible for breast conservation, the additional cancer detection rate by MRI was 6·2% (95% CI 4·1–9·3) and the MRI-associated false positive rate was 14·2% (10·7–18·4).[38] Despite the use of baseline MRI, the reoperation rate after breast-conserving surgery was high (26·3%). This rate is in line with the 20% and 27% reoperation rates respectively observed in the MRI and control groups of the randomised IRCIS trial (p=0·11).[39]

Overall, current evidence therefore does not support the added value of breast MRI in guiding surgery for DCIS.[40]

Given that modern screening technologies have increased overdiagnosis of DCIS, but we cannot yet identify which cases of DCIS are overdiagnosed, the metrics for evaluation of new screening methods should avoid tumour detection endpoints to limit the expansion of overdiagnosis (panel).[44] Refined pathological and biological assessments, and the incorporation of image analysis approaches should better inform decisions regarding the need for biopsy and, subsequently, the need for surgery. Attempts to apply artificial intelligence approaches to mammographic images in order to better predict progression to invasive disease are promising but still in development.[45]

Building informative prognostic and therapeutic biology

Since only one in six to ten untreated cases of DCIS might ultimately progress to invasive breast cancer, distinguishing indolent DCIS lesions from those with the potential for invasive progression is important.[8] However, invasive progression by itself is an imperfect surrogate of long-term outcomes, since many invasive cancers also display indolent behaviour and potentially represent overdiagnosis.[46] Most DCIS studies have used overall risk of recurrence and the risk of invasive recurrence as endpoints; future work should also subclassify invasive recurrence on the basis of biology and stage.

Pathologically, DCIS is graded as low (grade 1), intermediate (grade 2), or high (grade 3), on the basis of

Panel: Open questions and challenges

How can we decrease the incidence of screening-driven overdiagnosis of ductal carcinoma in situ (DCIS)?
Solutions to increase the specificity of breast cancer screening are currently being evaluated. These include stratified, risk-based approaches for screening, new imaging technologies, artificial intelligence-driven interpretation of images, and the development of blood-based detection or diagnosis biomarkers. Moreover, studies evaluating new screening tools should avoid pure, numerical tumour detection endpoints.

How can we better identify progression-prone (ie, high-risk) DCIS versus indolent, overdiagnosed (ie, low-risk) DCIS?
Ongoing and future work on the natural history of DCIS investigating prospective cohorts with modern multiomics, single cell methods, and artificial intelligence, among other methods, should ultimately provide reliable prognostic biomarkers and tools to sustain clinical decisions.

How can we better inform patients' decisions and generalise the use of decision aid tools?
Together with adequate education of health-care professionals, more collaborative and participative work is necessary to develop, validate, and widely spread the use of relevant, high-quality, and updated shared decision tools. Lay advocacy organisations are instrumental to this end.

Which patients require surgery and which could undergo active surveillance?
The results of the four ongoing prospective clinical trials assessing this question (table 1) should provide the first answers to this question. New prospective cohorts or randomised trials among women with newly diagnosed DCIS should be conducted to show the safety of further de-escalation among broader and better selected populations.

For whom should mastectomy still be proposed?
This decision is driven by the extent of DCIS on imaging or patient preference. Solutions must address both aspects. Large, indolent DCIS could be observed if characterised preoperatively through well validated molecular diagnostics and advanced imaging. Better patient education informed by new de-escalation trials would then define who should not have mastectomy.

Which patients benefit from post-operative radiation therapy after breast-conserving surgery?
Prospective trials (some of them ongoing) will show in which populations radiation therapy could be safely omitted, on the basis of baseline risk assessment with biomarkers and new follow-up imaging methods.

Who derives meaningful benefit from endocrine treatment after a DCIS diagnosis?
New biomarkers derived from our rapidly growing knowledge on DCIS progression mechanisms should help solve this issue. New options might be available among next generation endocrine agents, which could allow a better risk–benefit ratio.

Could we identify new preventive strategies to avoid the development of aggressive DCIS?
In the past 5 years, results from basic and translational research have opened new avenues for the development of biomarker-driven metabolic, immune, or targeted interventions.

Who derives meaningful benefit from adjuvant endocrine treatment after a DCIS diagnosis?
New biomarkers derived from our rapidly growing knowledge on DCIS progression mechanisms should help solve this issue. Functional assessments of endocrine benefits (such as short-term preoperative treatment exposure with biomarker-based response assessment) could be useful. New options might be available among next generation endocrine agents, which could allow a better risk–benefit ratio.

Could we identify new preventive strategies to avoid the development of aggressive DCIS?
In the past 2–5 years, results from basic and translational research have opened new avenues for the development of biomarker-driven interceptive metabolic, immune, or targeted interventions.

	COMET[41]	LORD[42]	LORETTA	LORIS[43]
Trial registration	NCT02926911	NCT02492607	UMIN000028298	ISRCTN27544579
Country	USA	Netherlands and Europe	Japan	UK
Design	RCT	Patient's preference	Single-arm trial	RCT
Main eligibility criteria	Older than 40 years with grade-1 or grade-2, pure DCIS; ER-positive ≥10%; HER2-negative	Aged 45 years or older with unilateral grade-1 or grade-2, pure DCIS; ER-positive ≥80%; HER2-negative	Aged 40 years or older with grade-1 or grade-2, pure DCIS; ER-positive; HER2-negative; lesion size ≤25 mm	Aged 46 years or older with grade-1 or grade-2, pure DCIS
Standard of care	Surgery, radiotherapy, endocrine therapy, or a combination	Surgery, radiotherapy, endocrine therapy, or a combination	Surgery, radiotherapy, endocrine therapy, or a combination	Surgery, radiotherapy, endocrine therapy, or a combination
Endocrine therapy allowed in active surveillance group?	Yes	No	Mandatory (tamoxifen)	Yes
Evaluation follow-up period	2, 5, and 7 years	10 years	5 and 10 years	10 years
Year opened	2017	2017	2017	2014
Target participant accrual (and actual when available)	1200 (997)	2500 (1250)	340	932 (181)
Planned end of accrual	2023	2025	2025	2020 (closed with 188 participants accrued)

DCIS=ductal carcinoma in situ. ER=oestrogen receptor. HER2-negative=human epidermal growth factor receptor is non-overexpressed and non-amplified. RCT=randomised controlled trial.

Table 1: **Overview of the ongoing DCIS clinical trials testing the safety of active surveillance for low-risk DCIS**

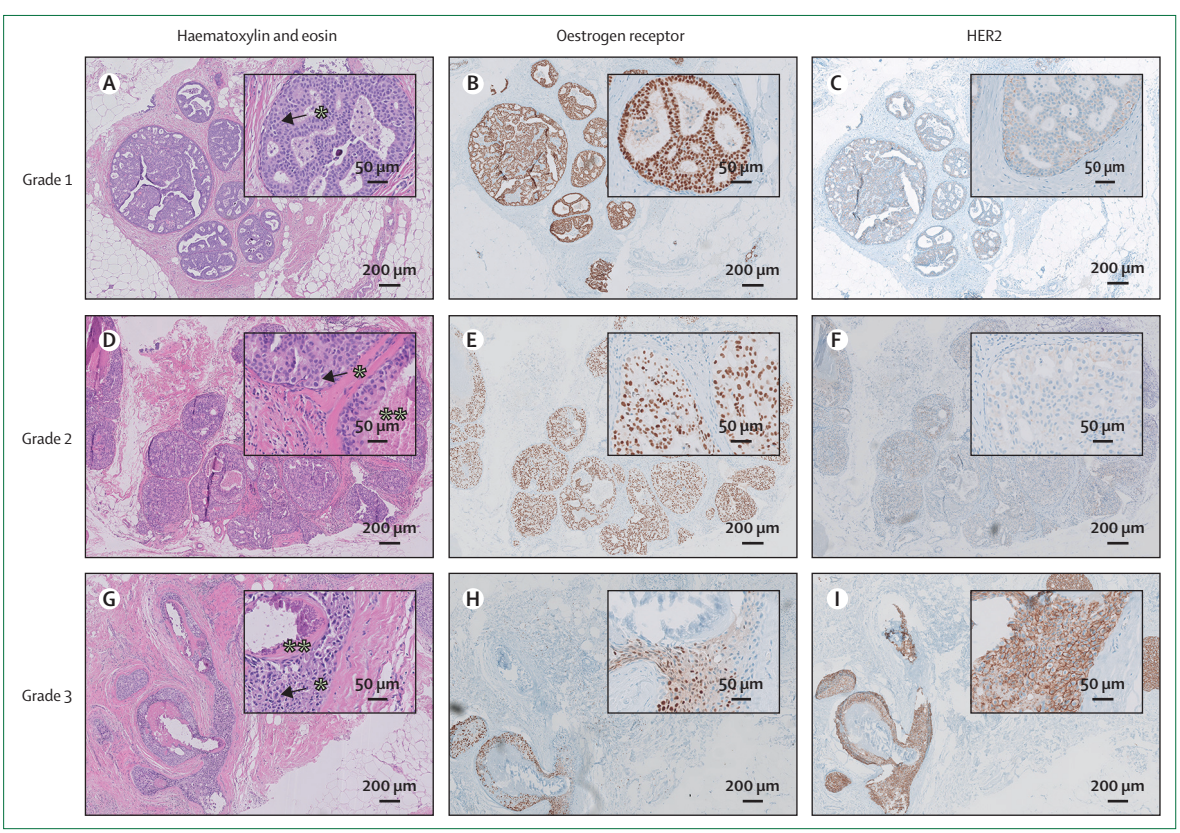

Figure 1: **Examples of low-grade, intermediate grade, and high-grade DCIS, often with calcifications**
(A–C) Low-grade DCIS (DCIS grade 1 lesion) with neoplastic luminal cells with small, monomorphic nuclei (asterisk) and almost no mitoses (see arrow pointing at the nucleus of neoplastic cell). (D–F) Intermediate grade DCIS (DCIS grade 2) with neoplastic luminal cells with moderately enlarged and irregularly shaped nuclei (asterisk), hard to find mitoses (arrow), and (sometimes) comedonecrosis (ie, necrosis in the lumen of the ducts; double asterisk). (G–I) High-grade DCIS (DCIS grade 3) with neoplastic luminal cells with marked, enlarged, irregular nuclei (asterisk), often easy to find mitoses (arrow), and (often) comedonecrosis (double asterisk). Size bars are 200 µm for overviews and 50 µm for inserts. Almost all DCIS grade 1 and 2 lesions stain positive for ER (darker staining in B and E, respectively) and approximately 10% are HER2-positive (darker staining in C and F, respectively), whereas in DCIS grade 3 lesions, approximately 50% are ER-negative (darker staining in H) and 50% are HER2-positive (darker staining in I).[47] DCIS=ductal carcinoma in situ. ER=oestrogen receptor. HER2=human epidermal growth factor receptor 2.

cytonuclear features (figure 1).[48] Extent and resection margins, described as prognostic in older studies, have retained weak predictive ability in contemporary series.[49] Grading by itself also has little value in predicting progression to invasive breast cancer and is subject to substantial interobserver variation.[47] Untreated high-grade DCIS seems to have a higher risk of subsequent invasive breast cancer than non-high-grade DCIS (17·6% *vs* 12·2% after 10 years of follow-up).[8] Overall, conflicting results have been obtained for the prognostic value of most markers tested (whether clinical or pathological), including patient age, grade, hormone receptors, human epidermal growth factor receptor 2 (HER2), tumour infiltrating lymphocytes (TILs), proliferation, and immune markers.[50] Traditional molecular invasive breast cancer classifiers do not perform well on DCIS.[50] Standalone immune markers cannot predict the development of invasive breast cancer.[51] High TILs added prognostic value to classic variables in the SweDCIS radiation therapy (RT) trial for HER2-negative lesions,[52] but this effect was small and inconsistent across studies. Concentrations of activated CD8 cells or the CD8–forkhead box P3 ratio might predict which cases of DCIS are likely to progress to invasive breast cancer.[53] The prevalence of HER2 amplification is 34% in DCIS and 13% in invasive breast cancer, but overall, this high prevalence is not associated with a higher risk of subsequent invasive breast cancer.[54]

No individual genomic features have yet been able to reproducibly distinguish DCIS with invasive and recurrence potential from DCIS without such potential.[15,20] To improve prognostication and guide management of DCIS in clinical practice, several molecular assays have been developed. The 12-gene DCIS score relies on the combined expression pattern of seven normalised genes to estimate the risk of local recurrence (invasive or non-invasive) after breast-conserving surgery.[55] The score has been shown to predict the risk of any local recurrence and invasive recurrence (hazard ratio [HR] 3·68 [95% CI 1·3–9·6] for the latter) following surgery alone.[55] A second example is the 7-protein immunohistochemistry score (also known as DCISionRT). This test combines centrally assessed cyclooxygenase-2, forkhead box protein A1, HER2, Ki-67, p16^{INK4A}, progesterone receptor, and seven in absentia homolog 2 immunohistochemistry expression with clinical factors (ie, age, size, palpability, and margin status). Initially developed as an independent predictor of invasive ipsilateral events (HR 3·1 [95% CI 1·5–6·5]) on two cohorts of patients with DCIS who received surgery or radiotherapy or both, this score has subsequently been independently and retrospectively validated on data from the SweDCIS randomised trial, which compared breast-conserving surgery only with breast-conserving surgery with radiotherapy.[56] Although these assays are appealing, they are not endorsed by most professional guidelines, and have several limitations. They are only useful for patients who receive breast-conserving surgery, are mostly prognostic with little predictive power regarding the benefit of radiotherapy, do not have strong prospective validation, and are expensive, requiring that samples be sent for central testing.[57,58] A number of ongoing studies (eg, NCT04797299, NCT03448926, and NCT04916808) are attempting to prospectively validate the clinical utility of these assays in patients who have undergone proper breast-conserving surgery. A new, promising 812-gene expression classifier has recently been identified that predicts local recurrence in the TBCRC 038 and RAHBT cohorts (HR 22·5 [95% CI 8·5–59·4] in the training set and 7·3 [1·6–34·2] in the validation set). They identified three transcriptomic DCIS subgroups, characterised by oestrogen receptor (ER) signalling, proliferation, and metabolism, and four stromal subtypes with distinct stromal and immune cell composition.[50] Further validation of these data is needed, however, particularly in view of the heterogeneity between the two cohorts, both in the surgical treatment used and the observed event rates.

Overall, although the existence of low-risk DCIS is widely acknowledged, available individual biomarkers cannot reliably define it. A few classifiers have been developed that can inform decision making post surgery only, but they do not have prospective validation.

Developing tailored surgical management

Until the 1980s, the standard treatment for DCIS was mastectomy. In the 1990s, several trials[59–62] established the appropriateness of breast-conserving surgery for many DCIS patients. These trials established two tenets that guide breast conservation for DCIS today: complete resection with free margins and the use of postoperative radiotherapy.[63] Microscopically tumour-free resection margins halved the risk of ipsilateral breast recurrence, regardless of the use of radiotherapy.[49,61] Prompted by continued uncertainty regarding optimal margin width for DCIS, a widely accepted consensus statement, based on a meta-analysis,[64] defined free margins as a 2 mm distance or more between tumour and ink.[65] Some heterogeneity persists in the application of this guideline across countries. New methods for intraoperative margin evaluation that could allow further minimisation of re-excision rates are under development, ranging from microCT to MRI, PET, or infrared imaging.[66]

Mastectomy continues to be used in 25–30% of patients with DCIS, either because it is indicated on the basis of the extent of their lesion, or because they opt for it, sometimes because of demographic and social factors.[67] A trend in the past 15 years towards the use of bilateral mastectomy, particularly among women younger than 50 years,[67,68] is partly explained by the increasing use of skin-sparing or nipple-sparing mastectomy, with its improved reconstruction results and better patient-reported outcomes.[69] The long-term rate of nipple

recurrence in the DCIS setting appears to be in the 2–3% range.[70] Reporting superficial margin status might be of importance in this regard.[71] In the past 5 years, explorations of single-port robotic mastectomy suggest improved patient satisfaction and similar complication rates,[72] but this procedure is still exploratory. Finally, with nodal positivity rates in the 3% range, pathological evaluation of axillary lymph nodes through sentinel node biopsy is not thought to be necessary for most individuals undergoing breast-conserving surgery. Current guidelines recommend this procedure if the DCIS is large or of high risk (eg, due to a high grade or the presence of a mass lesion), especially when treated with mastectomy or with excision in an anatomic location compromising the performance of a future sentinel node procedure.[36]

The major surgical question at present is whether excision can be omitted and replaced with active surveillance when the risk of invasive recurrence is considered low. The effect of omitting surgery has been assessed with the SEER data, in which 2% of 57 000 patients with DCIS treated between 1988 and 2011 had neither surgery nor radiotherapy.[8] With the limitation of the retrospective nature of these data, the 10-year breast cancer-specific survival rate was 98·5% following surgery, and 93·4% for those who did not have surgery, but was equal in patients with low-grade DCIS (98·8% following surgery and 98·6% for individuals who did not have surgery). This finding and those from other similar studies,[10] which identified low-risk patients mainly on the basis of grade and age (with variable cutoffs but generally identifyng an age of younger than 50 years as being high-risk),[73] have led to clinical trials that are evaluating the safety of active surveillance—avoiding any surgical resection for what is considered low-risk DCIS (ie, non-high-grade, ER-positive, and HER2-negative carcinomas).[41–43] Such an approach might spare many individuals with indolent DCIS the burden of surgery and radiotherapy and their associated side-effects. Four such trials have been initiated, three of which are ongoing (table 1). Although the randomised LORIS trial[43] stopped accrual in 2020 due to poor accrual, the new patient-preference design of the LORD trial[42] allowed a rapid increase in accrual. Initial barriers that were identified included physicians' and patients' fear of treatment de-escalation. The LORD trial is the only pure active surveillance trial, since it does not allow endocrine therapy in the active surveillance group. These trials will provide much-needed prospective evidence for the de-escalation of surgery for individuals with low-risk DCIS.

Radiotherapy of DCIS revisited

A proportion of patients with DCIS treated with breast-conserving surgery alone can develop local recurrence (up to 28% at 10 years post surgery in older trials), 50% of which will be invasive. In a recent Dutch population-based study,[74] 7·1% (95% CI 5·5–9·1) of invasive relapses occurred at 10 years. Multiple randomised trials have shown that radiotherapy to the whole breast reduces the risk of local recurrence, both invasive and non-invasive, by approximately 50%, but with no measurable effect on breast cancer-specific or all-cause mortality (table 2).[63] Data from the past 5 years, however, suggest that the treatment effect of radiotherapy on local recurrence might be limited to the first 10 years after surgery.[77,81] Radiotherapy to the breast has been associated with a 42–45% rate of acute grade 2 or higher toxicities, such as dermatitis and fatigue; and a 13–24% rate of late grade 2 or higher toxicities, such as breast pain, shrinkage, and induration, which can affect cosmesis and quality of life.[78,80] In addition, radiotherapy could cause rare (<0·5% incidence) serious side-effects, such as heart disease and second cancers, although such risk is expected to be much lower with modern techniques.

A major need is therefore to identify patients who are at low risk (typically less than 10% at 10 years post surgery) of local recurrence following breast-conserving surgery, and for whom radiotherapy would be expected to have a small effect and could be omitted. In three prospective studies that used only clinicopathological factors to define a low-risk category, the risk of local recurrence at 10 years was 10% or higher (table 2).[75–77] More recently, retrospective studies[55,56] have shown that the two RNA or protein expression classifiers previously discussed can, in combination with clinicopathological factors, identify groups of patients with risks of local recurrence that are less than 10% at 10 years, but these results require prospective validation. Beyond prognostic testing, there is also an increasing interest in identifying predictive molecular biomarkers to identify patients who are less likely to derive benefit from radiotherapy, but this work is ongoing.[56,82]

Escalation and de-escalation trials

Historically, patients who had breast-conserving surgery for DCIS were treated with conventionally fractionated radiotherapy of 50 Gy in 25 fractions over 5 weeks to the whole breast. The BIG 3–07/TROG 07·01 trial[78] evaluated the addition of boost radiation to the tumour bed following whole breast irradiation and the use of hypofractionation (a larger dose per fraction, and fewer total fractions) in moderate-risk to high-risk DCIS (table 2).[78] Boost radiation (16 Gy in 8 fractions over 1·5 weeks) reduced the risk of local recurrence at 5 years post surgery from 7·3% to 2·9%, but was associated with an increase in grade 2 or higher breast pain (10–14% incidence), induration (6–14%), and worse cosmesis (16% cosmetic deterioration at 3 years post surgery compared with 6% at 3 years without boost; table 2).[83] However, no difference in local recurrence was observed between conventional fractionation and hypofractionation (42·5 Gy in 16 fractions over 3 weeks). These results support the efficacy of boost radiation as an escalating strategy for patients at higher risk but they must be

	Start date	N	Design	Intervention	Local event rate (invasive or non-invasive)	Local event rate (invasive only)	Survival outcomes	Relevant adverse events
Meta-analysis of randomised trials assessing breast radiotherapy for DCIS after breast-conserving surgery								
EBCTCG[63]	-	3729	Meta-analysis of four randomised trials	Observation *vs* WBI	28·1% *vs* 12·9% at 10 years of follow-up, p<0·00001	NA	NA	No difference in any survival outcome, including breast cancer-specific mortality
Published trials testing the omission of radiotherapy after breast-conserving surgery for low-risk DCIS								
Dana Farber[75]	1995	158	Single-arm trial	Observation for low risk (lesion ≤2·5 cm, low grade or intermediate grade, margins ≥1 mm)	15·6% at 10 years of follow-up	NA	NA	NA
ECOG-ACRIN E5194[76]	1997	561	Single-arm trial	Observation for low risk (lesion ≤2·5 cm, low grade or intermediate grade, margins ≥3 mm)	14·4% at 12 years of follow-up	7·5% at 12 years of follow-up	NA	NA
ECOG-ACRIN E5194[76]	1997	104	Single-arm trial	Observation for low risk (lesion ≤1 cm, high grade, margins ≥3 mm)	24·6% at 12 years of follow-up	13·4% at 12 years of follow-up	NA	NA
NRG/RTOG 9804[77]	1998	636	Randomised	WBI *vs* observation for low risk (lesion ≤2·5 cm, low grade or intermediate grade, margins ≥3 mm)	7·1% *vs* 15·1% at 15 years of follow-up, p=0·007	5·4% *vs* 9·5% at 15 years of follow-up, p=0·03	NA	1% late, grade-3 toxicities (eg, pain, fibrosis, and breast retraction), no grade-4 or grade-5 toxitiy with radiotherapy
Randomised radiotherapy trials assessing hypofractionation, PBI, and boost radiation								
BIG3–07/TROG 07·01[78]	2007	1608	Randomised trial (4 arms)	WBI *vs* WBI and boost radiation (non-low-risk DCIS); conventional fractionation *vs* hypofractionation	7·3% *vs* 2·9% at 5 years, p<0·001; 5·1% *vs* 5·1% at 5 years of follow-up, p=0·85	NA	NA	Boost radiation was associated with an increased risk of any acute event (pain, swelling, radiodermatitis, etc [OR 2·1, 95% CI 1·3–3·5]) and any late event (1·8, 1·4–2·2), including pain, induration, and telangiectasia
NSABP B-39[79]	2005	1026	Randomised trial	WBI *vs* PBI in DCIS only subgroup	6·5% *vs* 6·0% at 10 years of follow-up	NA	NA	No differences observed when acute and late toxicities were combined
RAPID[80]	2006	381	Randomised trial	WBI *vs* PBI in DCIS only subgroup	3·7% *vs* 6·8% at 8·5 years of follow-up	NA	NA	Decreased acute toxicity, but increased late toxicity (32% *vs* 13% grade-2 toxicities or higher) and adverse cosmesis with PBI
Ongoing trials testing the omission of radiotherapy after breast-conserving surgery for low-risk DCIS								
ROMANCE (NCT03878342)	2019	666	Randomised trial	WBI *vs* observation for low risk (ER-positive or PR-positive, HER2-negative, Ki67 <15%, aged 50 years or older, lesion ≤2·5 cm)	··	··	··	··
ELISA (NCT04797299)	2021	526	Prospective cohort without a comparator group	Observation for low risk (Oncotype risk <10% at 10 years, older than 45 years, size ≤2·5 cm)	··	··	··	··

DCIS=ductal carcinoma in situ. ER=oestrogen receptor. HER2-negative=human epidermal growth factor receptor is non-overexpressed and non-amplified. NA=not assessed. OR=odds ratio. PBI=partial breast irradiation. PR=progesterone receptor. WBI=whole breast irradiation.

Table 2: **Overview of the design and results of the main DCIS radiotherapy trials**

weighed against the risk of increased toxicity. They also support the use of hypofractionation for patients receiving radiotherapy.

Partial breast irradiation (radiotherapy limited to the tumour bed) has been compared with whole breast irradiation in a number of trials that included a subgroup of patients with DCIS only (table 3).[80,83] Results have been mixed, and current guidelines support the use of partial breast irradiation for low-risk DCIS only.[90] Finally, a number of ongoing de-escalation trials (eg, NCT04797299 and NCT03878342) are exploring the omission of radiotherapy in molecular biomarker-identified, low-risk DCIS following breast-conserving surgery (table 2).

Systemic treatments: where do we stand?

More than two-thirds of all ductal carcinomas in situ express the ER protein.[9] International guidelines generally recommend testing DCIS for ER expression.[91] Five large, randomised trials have tested adjuvant endocrine therapy

	Criteria	N	Randomisation scheme	Follow-up period	5- year rate of any breast cancer event	Incidence of any in-breast event	All ipsilateral breast cancer events	Invasive ipsilateral breast cancer events	Contralateral breast cancer events	Major safety findings
NSABP B24[84,85]	Ductal carcinoma in situ; any ER status; lumpectomy and radiotherapy	1804	Tamoxifen 20 mg/day *vs* placebo for 5 years	74 months (57–93)	8·2% (tamoxifen) *vs* 13·4% (placebo)	0·63 (0·47–0·83)	0·70 (0·50–0·98)	0·56 (0·32–0·95)	0·48 (0·26–0·87)	Increased incidence of endometrial cancer with tamoxifen (1·53 *vs* 0·45 per 1000 participants per year); no increase in thromboembolic events
UK/ANZ[86]	Ductal carcinoma in situ; any ER status; lumpectomy with or without radiotherapy	1701	Tamoxifen 20 mg/day *vs* adjuvant endocrine treatment for 5 years	12·7 years (10·9–14·7)	14% (tamoxifen) *vs* 18% (none)	0·71 (0·58–0·88)	0·78 (0·62–0·99)	0·95 (0·66–1·38)	0·44 (0·25–0·77)	NR
TAM-01[87]	70% ER-positive DCIS	500	Tamoxifen 5 mg/day *vs* adjuvant endocrine treatment for 3 years	5·1 years (3·9–6·3)	6·4% (tamoxifen) *vs* 11% (placebo)	0·48 (0·26–0·92)	NR	NR	0·25 (0·07–0·88)	Slightly more hot flashes (14% *vs* 7%) and vaginal bleeding (4% *vs* 1·2%) with tamoxifen than placebo; no increase in thromboembolic events or endometrial cancer
IBIS2[88]	Post-menopausal; ER-positive DCIS; lumpectomy with or without radiotherapy	2980	Anastrozole 1 mg per day *vs* tamoxifen 20 mg per day for 5 years	7·2 years (5·6–8·9)	2·5% (anastrozole) *vs* 3·0% (tamoxifen)	0·89 (0·64–1·23)	NR	0·77 (0·40–1·48)	NR	Decreased incidence of endometrial cancer with anastrozole (1 event *vs* 11 events); increased incidence of fractures (9% *vs* 7%), musculoskeletal events (64% *vs* 54%), and hypercholesterolaemia (3% *vs* 1%) with anastrozole; decreased hot flashes (56% *vs* 60%) and thromboembolic events (<1% *vs* 2%) with anastrozole
NBSABP B35[89]	Post-menopausal; ER-positive DCIS; lumpectomy and radiotherapy	3104	Anastrozole 1 mg per day *vs* tamoxifen 20 mg per day for 5 years	9 years (8·2–10)	8·5% (anastrozole) *vs* 8·4% (tamoxifen)	0·73 (0·56–0·96); effect limited to women younger than 60 years	0·83 (0·56–1·22)	0·76 (0·40–1·43)	0·64 (0·43–0·96)	Decreased incidence of endometrial cancer (8 events *vs* 17 events) and thromboembolic events (0·8% *vs* 2·7%) with anastrozole; increased incidence of fractures (69 events *vs* 50 events), arthralgia (32·8% *vs* 23·3% grade ≥2) with anastrozole

Data are median (95% CI), rate, or HR (95% CI). DCIS=ductal carcinoma in situ. ER=oestrogen receptor. HR=hazard ratio. NR=not reported.

Table 3: **Overview of the efficacy results of the five randomised trials assessing the benefits of adjuvant endocrine treatment in DCIS**

among women who have had breast conservation surgery with and without radiotherapy in the past 25 years (table 3). Treatment with tamoxifen for 5 years was investigated in the National Surgical Adjuvant Breast and Bowel Project (NSABP) B-24 trial[84,85] and the UK, Australia, and New Zealand DCIS trial.[60,86] In B-24, at 6 years of follow-up, a 37% reduction of all new breast cancer events was observed with tamoxifen compared with placebo.[84] A post-hoc analysis showed that this benefit was limited to ER-positive DCIS, for which the risk reduction was 51% at 10 years.[85] In the UK, Australia, and New Zealand trial, at 12·7 years of follow-up, tamoxifen reduced the incidence of all new breast events by 29% (30% for ipsilateral DCIS and 66% for contralateral events).[60] However, no effect was detected on ipsilateral invasive disease (HR 0·95, 95% CI 0·66–1·38; p=0·8). A more recent randomised, placebo-controlled trial (N=500) published in 2023 showed that a low dose (5 mg) of tamoxifen in a mixed population, 70% of which was patients treated for DCIS, halved the incidence of breast-invasive or DCIS events in the long term.[87] Two randomised trials, NSABP B35[89] and IBIS2,[88] evaluated the effect of an aromatase inhibitor, anastrozole, versus tamoxifen in patients with ER-positive DCIS treated with breast conservation surgery and radiotherapy. Although IBIS2 only showed non-inferiority to tamoxifen,[88] NSABP B35 showed a 27% relative reduction in breast cancer events with anastrozole compared with tamoxifen; however, this superiority was only observed in

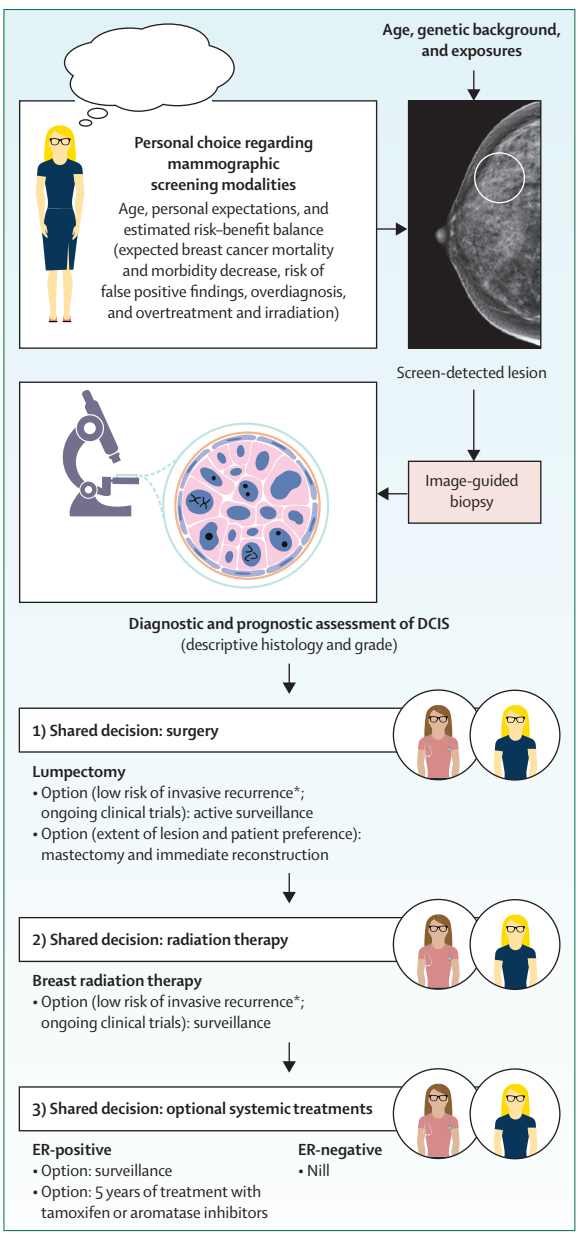

Figure 2: **Overview of finding the balance between overtreatment and undertreatment of DCIS of the breast**

DCIS=ductal carcinoma in situ. ER=oestrogen receptor. HER2=human epidermal growth factor receptor 2. *For active surveillance trials, low risk is defined as pure, asymptomatic, screen-detected grade-1 or grade-2 and, for one of these trials, ER-positive and HER2-negative DCIS. Post-surgically, low risk might be estimated by bioclinical tests (but not in the context of active surveillance trials).

women younger than 60 years.[89] These efficacy results are supported by data from real-world populations.[9,92] In a large cohort of more than 35 000 women treated for screen-detected DCIS in the National Health Service (NHS), the cumulative 15-year rate of ipsilateral invasive breast cancer decreased by 36% in women with ER-positive disease who had endocrine treatment compared with those who did not (4·7% *vs* 7·3%).[9]

However, in a recent international cohort of more than 47 000 women,[49] endocrine therapy did not significantly reduce the risk of invasive ipsilateral events in women who had breast-conserving surgery for DCIS (HR 0·86, 95% CI 0·62–1·21).[49] These mixed findings need to be balanced against several issues. First, these endocrine treatments have no effect on survival outcomes. In the NHS cohort, the adjusted HR for death did not support endocrine therapy (HR 1·33, 95% CI 0·72–2·49).[9] This finding supports the view that endocrine therapy for DCIS is for prevention and is not an adjuvant therapy.[93,94] Second, the absolute expected benefit for an individual in terms of new breast events is approximately 5% (table 3), whereas the number needed to avoid one cancer is around 15–22%. The third issue is toxicity and quality of life, which must be weighed against the estimated benefits of the treatment. Besides negative effects on quality of life,[95] tamoxifen increases thromboembolic events and the risk of endometrial cancer, and aromatase inhibitors are associated with well known musculoskeletal, gynaecological, and metabolic side-effects (the most frequent being chronic arthralgia and myalgia, bone loss, vulvovaginal dryness, and dyslipidaemia).[88,89] The global uptake of prescribed adjuvant endocrine therapy for DCIS, whether this is tamoxifen or aromatase inhibitors, currently appears to be low, in the range of 12% or less in European real-world contexts.[9,96] Other endocrine approaches beyond low-dose tamoxifen are currently under investigation, including the trans-dermal delivery of anti-oestrogens (NCT02993159 and NCT04009044), but need further refinement. Another attractive approach for the approximate third of patients with HER2-positive DCIS involves targeting HER2.

Several studies have tested preoperative trastuzumab, which had suggested effects on tumour volume and proliferation.[97] In the NSABP B43 trial,[98] 2014 women with HER2-positive DCIS were randomly assigned following breast conservation surgery to radiotherapy with or without short-term concomitant trastuzumab. This approach reported a non-significant reduction in ipsilateral events of 19% in the trastuzumad group.[98] Overall, only endocrine therapy is a systemic treatment option for DCIS. It should be seen as a preventive strategy, with carefully weighted risks and benefits, for individuals with ER-positive DCIS.

Person-centred, risk–benefit approaches

Patients' journeys to informed treatment choices

The first important decision in patients' journeys to informed treatment strategies (figure 2) concerns their informed choice to receive breast cancer screening. All individuals, when advised to consider breast cancer screening, should be informed in the most personalised manner possible of the potential benefits and harms of such screening. Mean expected benefits for women include a decrease in breast cancer mortality of approximately 20% and a decrease in stage-related treatment needs.[46] Potential harms include false positive

findings possibly requiring biopsies for 5–10% of women over a 10-year screening period, depending on their age and breast density, and long-term effects of repeated irradiation, including a very small potential increase of the risk of breast cancer.[46] Finally, and importantly, there is an age-dependent risk of overdiagnosis and overtreatment (10–15% of invasive breast cancers and 3–65% of DCIS).[46] Several online decision tools that transform this information into easily understandable, graphically represented numbers are available globally to help women in their screening choice.[99]

Following a diagnosis of DCIS, the most trying decisions relate to local therapy (figure 2). The current local options, outside of a clinical trial, range from breast-conserving surgery with or without postoperative radiotherapy to mastectomy with reconstruction, since active surveillance is not currently a validated option. As summarised earlier, the benefits of the more aggressive options are higher local control rates, but these do not affect survival outcomes. Their risks include short-term and long-term surgical complications, pain, cosmetic changes and, in rare cases, more serious radiation-induced conditions such as second cancers. Anxiety affects a substantial proportion of women, up to a third in some studies,[100] and is highest in the period following diagnosis. Sexuality and body image are also affected by surgical treatment, and unsurprisingly, women undergoing mastectomy have reported the highest rates of dissatisfaction in this domain.[101] For individuals with hormone receptor-positive DCIS, several years of endocrine therapy might also be proposed, resulting in a 40–50% relative reduction in breast events, but at a cost of impaired patient-reported quality of life. Importantly however, the initial negative effect of treatments for DCIS on patients' physical wellbeing tends to diminish over time.[100]

Patients' perception of risk and shared decision making

Physicians must carefully estimate and communicate with their patients in a balanced manner their individual expected benefits and risks regarding each treatment step based on available knowledge, and guide them to make choices that suit their personal characteristics, priorities, and circumstances (eg, age, income, and social support). This process is foundational to the principle of shared decision making. It allows patients to be informed, to express their preferences, and to participate in decision making regarding the extent of surgery, the use of radiotherapy, and possibly the use of endocrine agents. Since the benefits of each individual approach overlap to some degree but are also distinct (eg, regarding ipsilateral *vs* contralateral events), specific information regarding these distinctions should also be provided to patients.

Challenges for physician–patient communication include explaining the difference between invasive and non-invasive disease, that DCIS is not a threat to wellbeing by itself, and that treatment recommendations are aimed at preventing future invasive breast cancer. Useful questions to ask the patient include their risk tolerance, the value they place on cosmetic appearance, the importance of logistical aspects such as geographical access to treatment centres, and their comorbidities (including bone health and risk factors for thromboembolism). Unfortunately, although they are readily available in many countries,[99] methods to improve communication, such as booklets, illustrations, and audiotaping of consultations, are often not routinely employed in clinical practice.[102] The fact that women with DCIS have reported high levels of dissatisfaction with communication and shared decision making for different treatment options is therefore not surprising.[101] However, in a recent publication of the patient preference-based LORD trial,[103] 75% of women opted for active surveillance of their low-risk DCIS versus 25% who opted for surgery. Increasing efforts need to be made to provide women with easy-to-understand, up-to-date information and decisional support for various treatment options.

Conclusion

Ductal carcinoma in situ is a multifaceted entity in oncology, being alternatively a precursor lesion that could lead to aggressive invasive cancer or an overdiagnosed and overtreated lesion that never progresses. Considerable knowledge has been accumulated in the past 10 years, improving our ability to start de-escalating the treatment of this entity in selected patients. High-quality translational and clinical research on one side and carefully supported shared decision making on the other are necessary approaches to improve lives, spare breasts, and enhance quality of life among many thousands of individuals each year.

Contributors

SD conceptualised this Review with all other authors. All authors were involved in conducting the literature search and in interpreting the literature. SD coordinated this Review and the writing of the manuscript; all authors contributed equally to the final writing of the manuscript. SAK was specifically in charge of the imaging and surgical parts of the manuscript; JW was in charge of the pathology, biology and biomarkers; TW was in charge of the radiotherapy and patient pathway; SD was in charge of the introduction, summary, systemic treatments, and conclusion. SD was also responsible for modifications to the manuscript based on reviewers' comments, and the homogenisation of the manuscript. SD coordinated the revision and final writing of the revised version of the manuscript. Figure 1 was provided by JW. Figure 2 was drafted by SAK and contributed to by all authors. This Review was edited and proofread several times by all authors, and all authors agreed on the final submitted version.

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